Linear Regression

Linear Regression

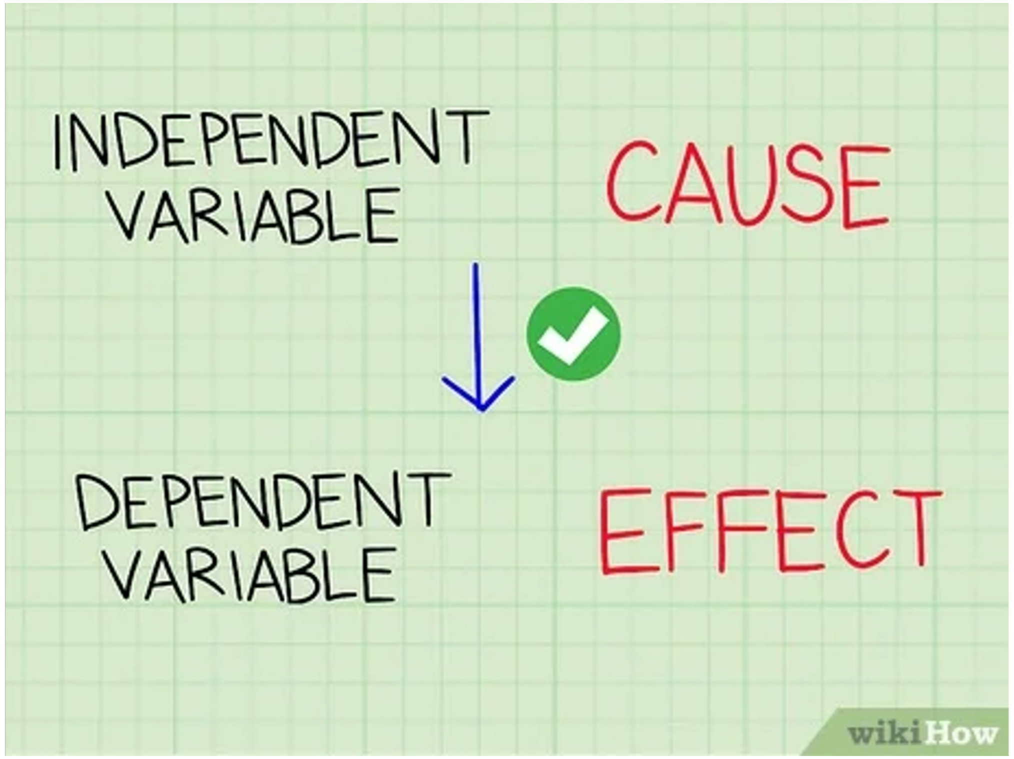
- Linear regression is a **supervised learning algorithm** used for modeling the relationship between a dependent variable (**Y**) and one or more independent variables (**X**).

- It assumes a linear relationship between the variables.

- A linear regression is a data plot that **graphs** the **linear relationship between an independent and a dependent variable(s).**

Linear Regression

- The variable you **want to predict** is called the **dependent variable.** The variable you are **using to predict** the **other variable's value** is called the **independent variable**



Dependent and Independent Variable

Predict a student's **exam score** based on the **number of study hours**

Independent Variable

Dependent Variable

Study Hours (X)	Exam Score (Y)
1	50
2	55
3	65
4	70
5	75

Linear Regression

Regression analysis is the process of estimating a functional relationship between X and Y. A regression equation is often used to predict a value of Y for a given value of X

- Y: is referred to as the *dependent variable*, the *response variable* or the *predicted variable*.
- X: is referred to as the *independent variable*, the *explanatory variable* or the *predictor variable*.

Types of Linear Regression

Simple Linear Regression

One independent variable (X) and one dependent variable (Y).

Multiple Linear Regression

More than one independent variable (X1, X2, X3, ... Xn).

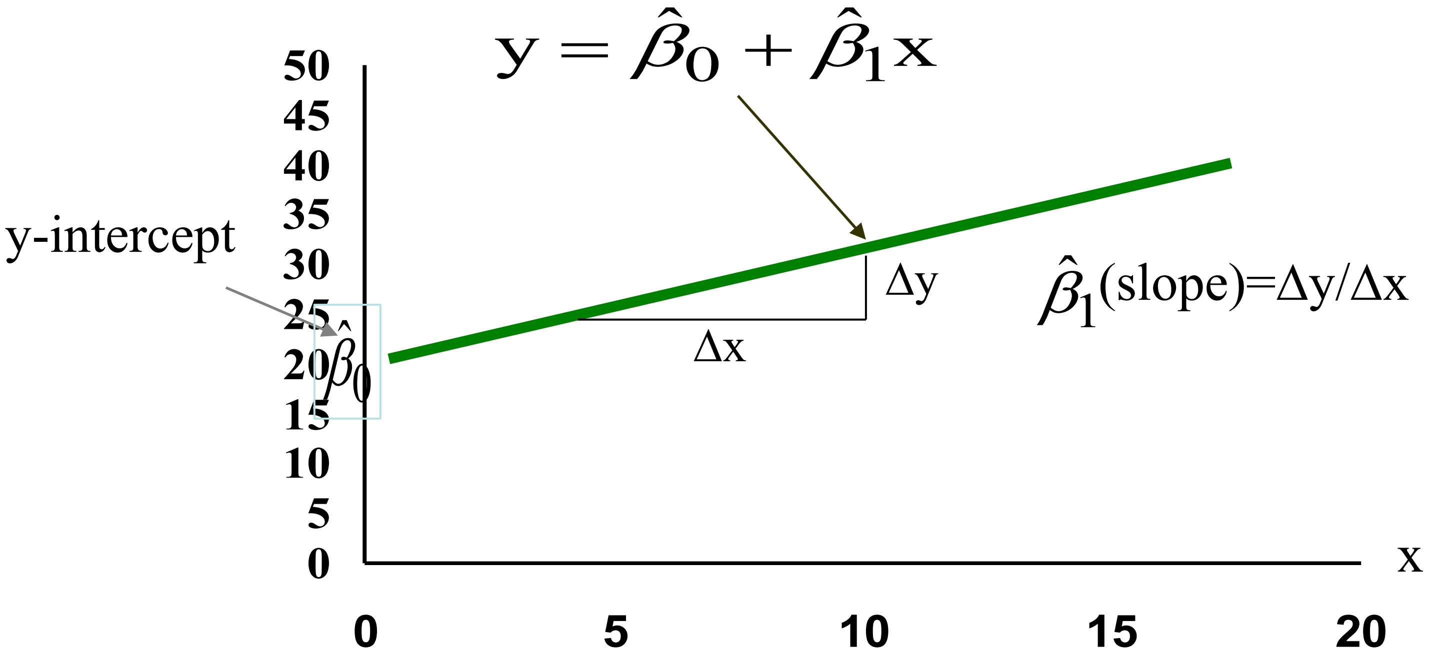
Linear Regression : Single Variable

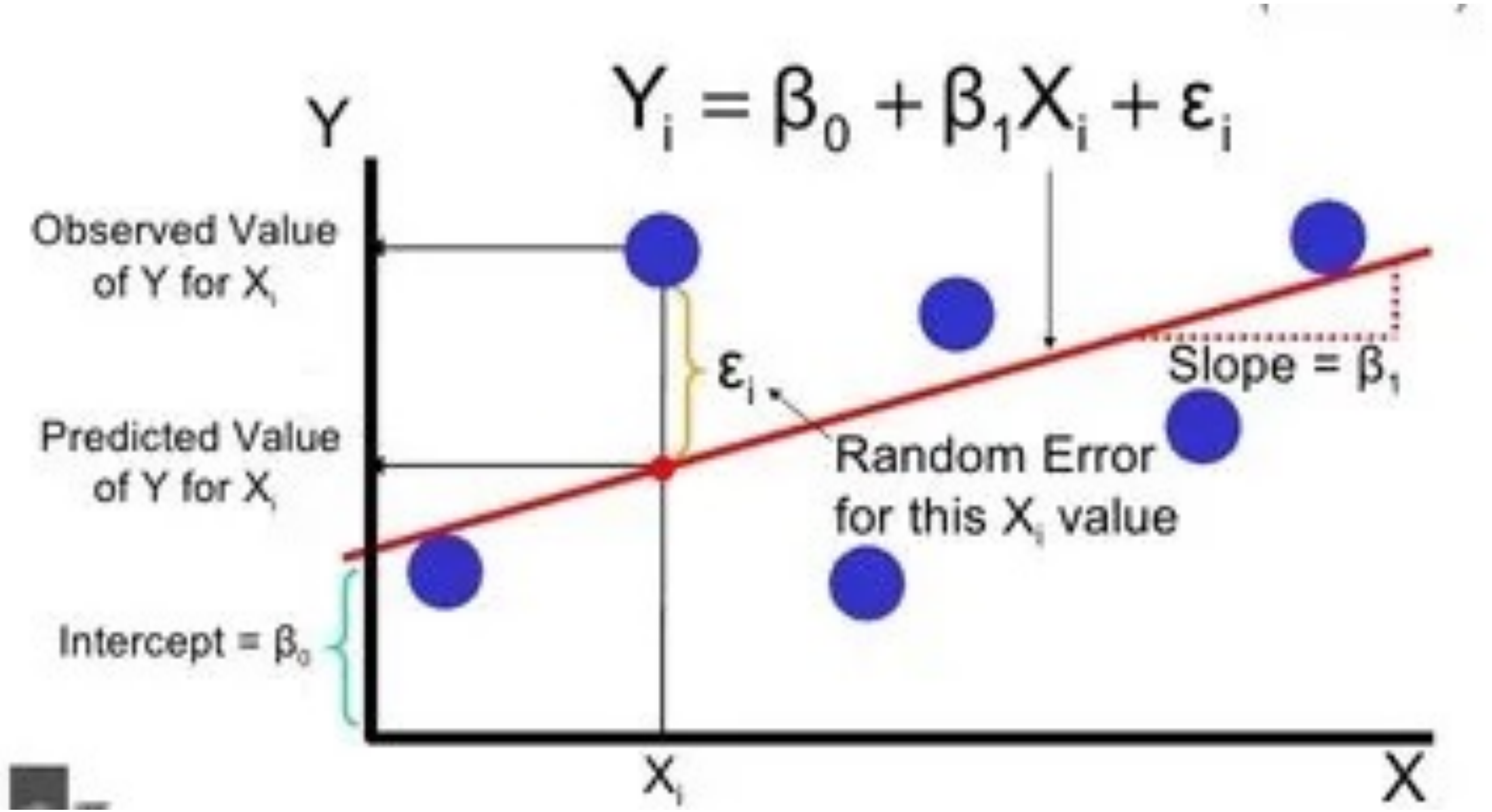
Linear Regression: Single Variable

$$\underset{\text{Predicted output}}{\boxed{\hat{y}}} = \underbrace{\beta_0 + \beta_1}_{\text{Coefficients}} \underset{\text{Input}}{\boxed{x}} + \underset{\text{Error}}{\boxed{\epsilon}}$$

Where,
Y = Dependent variable (Target)
X = Independent variable (Feature)
$\beta 0$ = Intercept (Constant term)
$\beta 1$ = Slope (Coefficient)
ε = Error term (Difference between predicted and actual value)

Deterministic Component of Model




$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$
Y
Observed Value of Y for X_i
ε_i
Predicted Value of Y for X_i
Random Error for this X_i value
Slope = β_1
Intercept = β_0
X_i
X

Single Linear Regression : Problem

- Let's us consider an example where the five weeks sales data in (Thousands) is given as below. Apply linear regression technique to predict the 7^{th} and 9^{th} month sales

Week (X)	Sales (Y) in Thousands
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Compute Slope (β1)Using Different Formulas

1. Using Mean-Based Formula

$$\beta_1 = \frac{\overline{XY} - \bar{X}\bar{Y}}{\overline{X^2} - (\bar{X})^2}$$

2. Using OLS Summation Formula

$$\beta_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$$

3. Using Standard Least Squares Formula

$$\beta_1 = \frac{\sum X_i Y_i - n\bar{X}\bar{Y}}{\sum X_i^2 - n\bar{X}^2}$$

Compute Intercept (β0)

$$\beta_0 = \bar{Y} - \beta_1\bar{X}$$

Compute Slope (β1)Using Mean Based Formula

x_i	y_i	$(x_i)^2$	$x_i \times y_i$
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19
Sum = 15 Average of (x_i) $= \bar{x} = \frac{15}{5}$ $= 3$	Sum = 12.6 Average of (y_i) $= \bar{y} = \frac{12.6}{5}$ $= 2.52$	Sum = 55 Average of (x_i^2) $= \overline{x_i^2} = \frac{55}{5}$ $= 11$	Sum = 44.4 Average of $(x_i \times y_i)$ $= \overline{xy} = \frac{44.4}{5}$ $= 8.88$

x_i	y_i	$(x_i)^2$	$x_i \times y_i$
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19
Sum = 15 Average of (x_i) $= \bar{x} = \frac{15}{5}$ $= 3$	Sum = 12.6 Average of (y_i) $= \bar{y} = \frac{12.6}{5}$ $= 2.52$	Sum = 55 Average of (x_i^2) $= \overline{x_i^2} = \frac{55}{5}$ $= 11$	Sum = 44.4 Average of $(x_i \times y_i)$ $= \overline{xy} = \frac{44.4}{5}$ $= 8.88$

$$\beta_1 = \frac{\overline{XY} - \bar{X}\bar{Y}}{\overline{X^2} - (\bar{X})^2}$$

$$\beta_1 = \frac{8.88 - (3 \times 2.52)}{11 - (3^2)}$$

$$= \frac{8.88 - 7.56}{11 - 9} = \frac{1.32}{2} = 0.66$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\beta_0 = 2.52 - (0.66 \times 3)$$

$$= 2.52 - 1.98 = 0.54$$

The Fitted line for above dataset is

$$\hat{Y} = 0.54 + 0.66X$$

The predicted 7th week sale would be (when $x = 7$), $y = 0.54 + 0.66 \times 7 = 5.16$ and the 12th month, $y = 0.54 + 0.66 \times 12 = 8.46$. All sales are in thousands.

Finding the Best Fit Line (Ordinary Least Squares Method - OLS)

- The best fit line minimizes the **sum of squared errors (SSE)**:

$$SSE = \sum (Y_i - \hat{Y}_i)^2$$

Where:

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value

The coefficients (β_0, β_1) are found using:

$$\beta_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\sum(X_i - \bar{X})(Y_i - \bar{Y})$$

X_i	$X_i - \bar{X}$	Y_i	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	-2	1.2	-1.32	$(-2) \times (-1.32) = 2.64$
2	-1	1.8	-0.72	$(-1) \times (-0.72) = 0.72$
3	0	2.6	0.08	$(0) \times (0.08) = 0.00$
4	1	3.2	0.68	$(1) \times (0.68) = 0.68$
5	2	3.8	1.28	$(2) \times (1.28) = 2.56$

$$\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 2.64 + 0.72 + 0.00 + 0.68 + 2.56 = 6.6$$

$$\sum(X_i - \bar{X})^2$$

X_i	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
1	-2	4
2	-1	1
3	0	0
4	1	1
5	2	4

$$\sum(X_i - \bar{X})^2 = 4 + 1 + 0 + 1 + 4 = 10$$

$$\beta_1 = \frac{6.6}{10} = 0.66$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$\beta_0 = 2.52 - (0.66 \times 3)$$

$$= 2.52 - 1.98 = 0.54$$

The Fitted line for above dataset is

$$\hat{Y} = 0.54 + 0.66X$$

A company wants to predict sales based on advertising expenses. Given the dataset

Advertisement Spend (X)	Sales (Y)
10	25
15	30
20	32
25	35
30	40

$$\hat{Y} = 18.4 + 0.7X$$

Error Metrics

- The following key metrics help measure its accuracy and performance. The given below error metrics measure how far the predicted values are from the actual values.

Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- Measures the average squared difference between actual and predicted values.
- Lower values indicate a better fit.

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

- The square root of MSE, making it easier to interpret since it has the same units as the target variable.

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

- Measures the average absolute difference between actual and predicted values.
- Less sensitive to outliers compared to MSE.

Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{n}\sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

- Expresses the error as a percentage of actual values.
- Used when the target variable is strictly positive.

Uses of Regression

Determining the strength of predictors

- Relationship between Strength of sales and marketing spending
- Relationship between age and income

Forecasting an effect

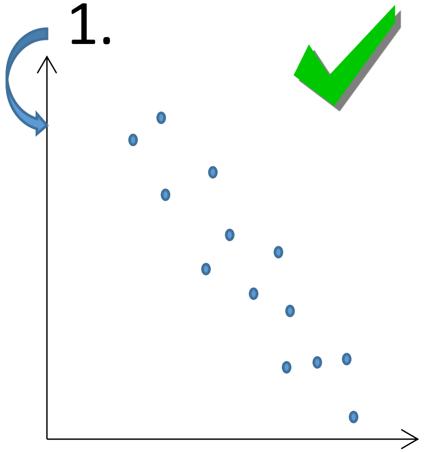
- How much additional sale income will I get for each Thousands of Rupee spent on marketing

Trend Forecasting /Point Estimates

- What will be the price of GOLD in next six months

Which of the following data are likely to be most appropriately modelled using a linear regression model?

1.



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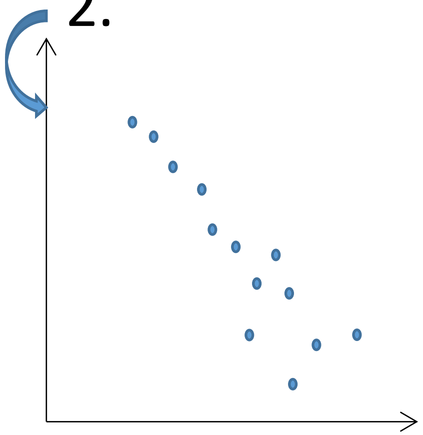

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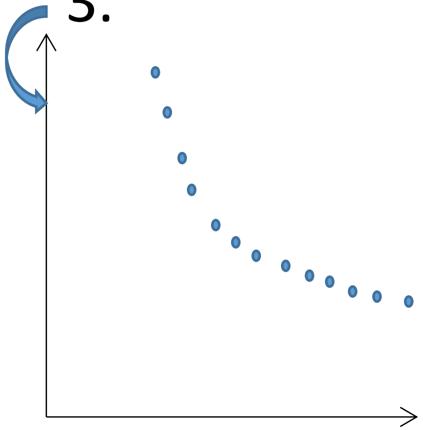
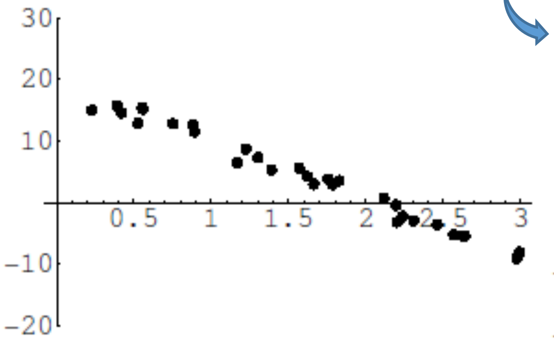


Which of the following plots would have the greatest residual sum of squares [variance of y about the fitted line]?

Question from Derek Bruff

1.



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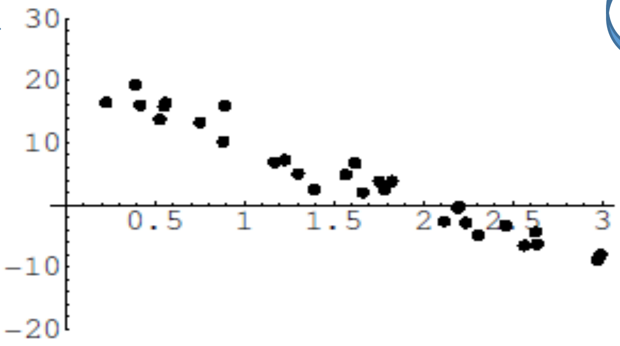

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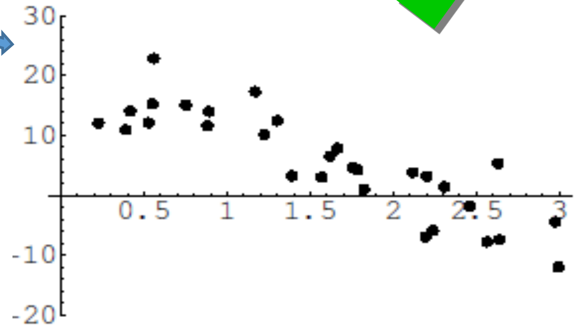
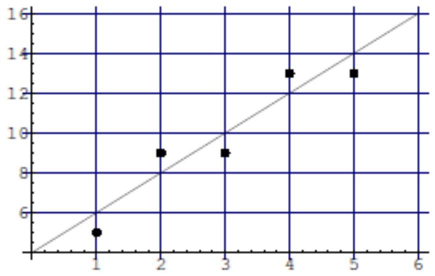


The line $y = 4 + 2x$ has been proposed as a line of best fit for the following four sets of data. For which data set is this line the best fit (minimum $E = \sum_i e_i^2$)?

Question from Derek Bruff

1.



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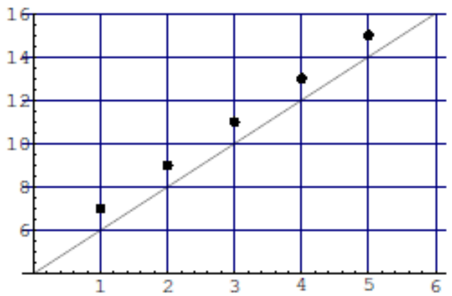

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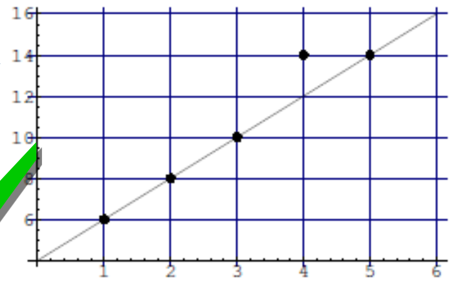

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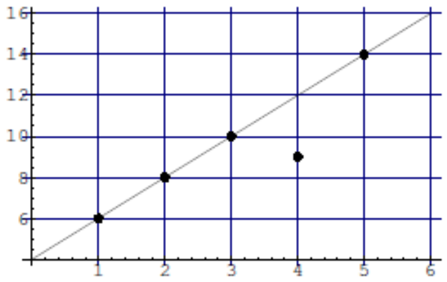


Multiple Linear Regression

- **Multiple Linear Regression** is an extension of simple linear regression where multiple independent variables (X1,X2,X3,.....) are used to predict a dependent variable (Y).
- Regression model for k independent variables:

$$\boxed{Y_i = a + b_1 X_{1i} + b_2 X_{2i} \ldots + b_k X_{ki} + e_i}$$

Example of Multiple Linear Regression

- Let's consider an example where we predict the **house price** (Y) based on three independent variables:
- **Size (sq ft)** (X1)
- **Number of Bedrooms** (X2)
- **Age of the House (years)** (X3)

Equation for Multiple Linear Regression

$$Price = \beta 0 + \beta 1(Size) + \beta 2(Bedrooms) + \beta 3(Age) + \epsilon$$

Least Squares Method (Normal Equation)

The best-fit line minimizes the **sum of squared errors (SSE)**:

$$\mathbf{B} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Steps to Compute Using the Normal Equation : Matrix Form

- **X** is the matrix of input features (including a column of 1s for the intercept).
- **y** is the target variable (dependent variable).
- **B** is the vector of regression coefficients (β0,β1,β2,...).

Solve for $\mathbf{B}$ using:

- Compute $\mathbf{X}^T\mathbf{X}$
- Compute its inverse $(\mathbf{X}^T\mathbf{X})^{-1}$
- Multiply by $\mathbf{X}^T\mathbf{y}$ to get $\mathbf{B}$

Let's consider a dataset where we want to predict **salary** based on:

- **Years of Experience** (X_1)
- **Number of Certifications** (X_2)

Years of Experience (X_1)	Certifications (X_2)	Salary (Y) ($1000s)
1	1	40
2	1	45
3	2	50
4	2	55
5	3	60

$$Y=\beta_0+\beta_1 X1+\beta_2 X2$$

where:

- β_0 = Intercept
- β_1= Coefficient for **Years of Experience**
- β_2= Coefficient for **Certifications**

Constructing Matrices

The feature matrix X (without the intercept):

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 2 \\ 4 & 2 \\ 5 & 3 \end{bmatrix}$$

The target variable vector y:

$$y = \begin{bmatrix} 40 \\ 45 \\ 50 \\ 55 \\ 60 \end{bmatrix}$$

Adding a column of ones to X for the intercept:

$$X_b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 3 & 2 \\ 1 & 4 & 2 \\ 1 & 5 & 3 \end{bmatrix}$$

Compute $\mathbf{X}^T\mathbf{X}$

$$\mathbf{X}^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 2 & 2 & 3 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 3 & 2 \\ 1 & 4 & 2 \\ 1 & 5 & 3 \end{bmatrix}$$

Compute $(\mathbf{X}^T\mathbf{X})^{-1}$

We have:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 5 & 15 & 9 \\ 15 & 55 & 32 \\ 9 & 32 & 19 \end{bmatrix}$$

Now, we compute its inverse:

$$(\mathbf{X}^T\mathbf{X})^{-1} = \begin{bmatrix} 1.16 & -0.6 & -0.4 \\ -0.6 & 0.4 & 0.0 \\ -0.4 & 0.0 & 0.8 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 5 & 15 & 9 \\ 15 & 55 & 32 \\ 9 & 32 & 19 \end{bmatrix}$$

Compute $\mathbf{X}^T\mathbf{Y}$

Given:

$$\mathbf{Y} = \begin{bmatrix} 40 \\ 45 \\ 50 \\ 55 \\ 60 \end{bmatrix}$$

Now, compute:

$$\mathbf{X}^T\mathbf{Y} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 40 \\ 45 \\ 50 \\ 55 \\ 60 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{Y} = \begin{bmatrix} 250 \\ 550 \\ 295 \end{bmatrix}$$

Compute β

We now compute:

$$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$$

$$\begin{bmatrix} 1.16 & -0.6 & -0.4 \\ -0.6 & 0.4 & 0.0 \\ -0.4 & 0.0 & 0.8 \end{bmatrix} \begin{bmatrix} 250 \\ 550 \\ 295 \end{bmatrix}$$

$$\boldsymbol{\beta} = \begin{bmatrix} 30 \\ 5 \\ 5 \end{bmatrix}$$

Final Regression Equation

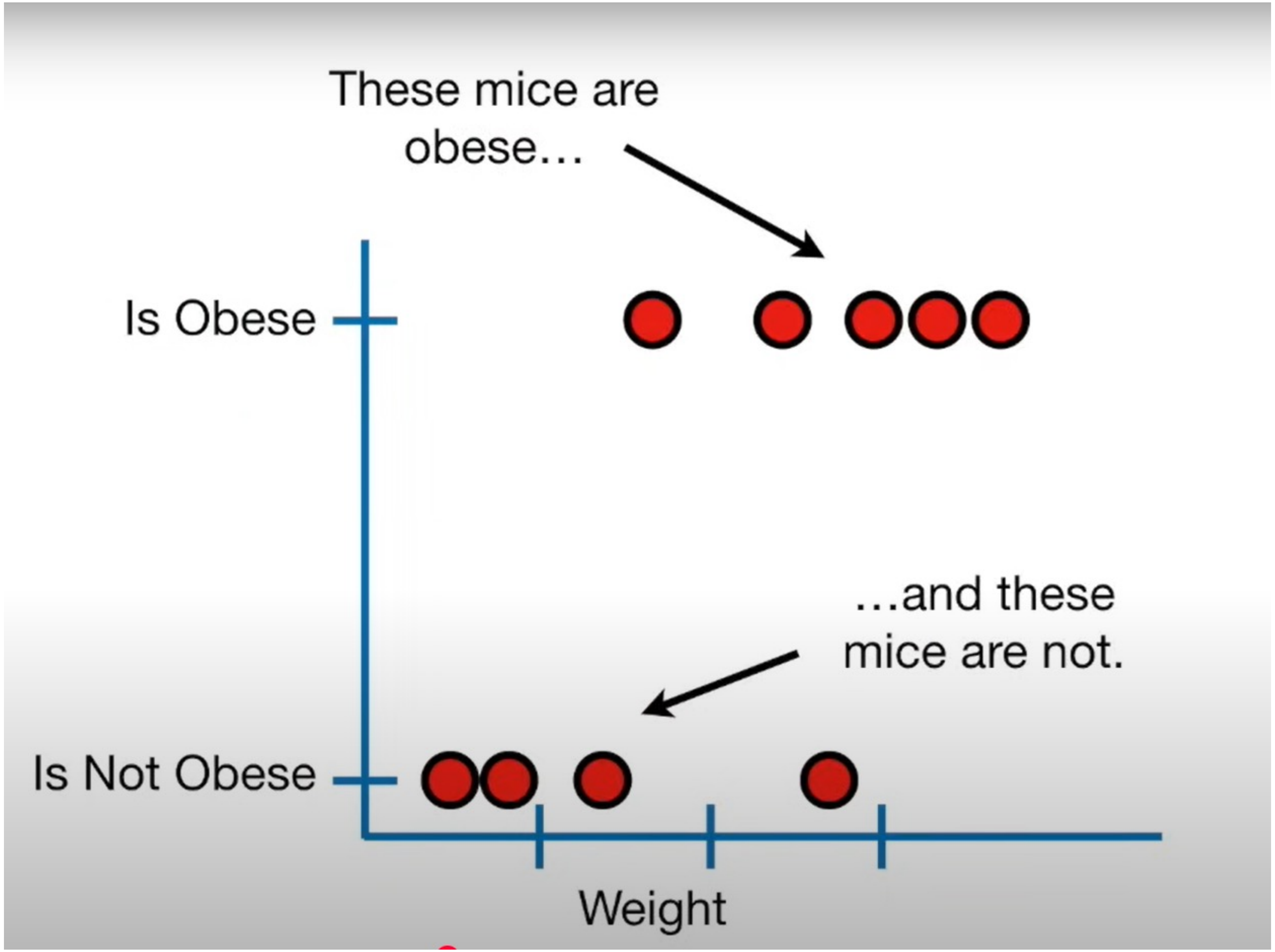
$$Y = 30 + 5X_1 + 5X_2$$

Logistic regression

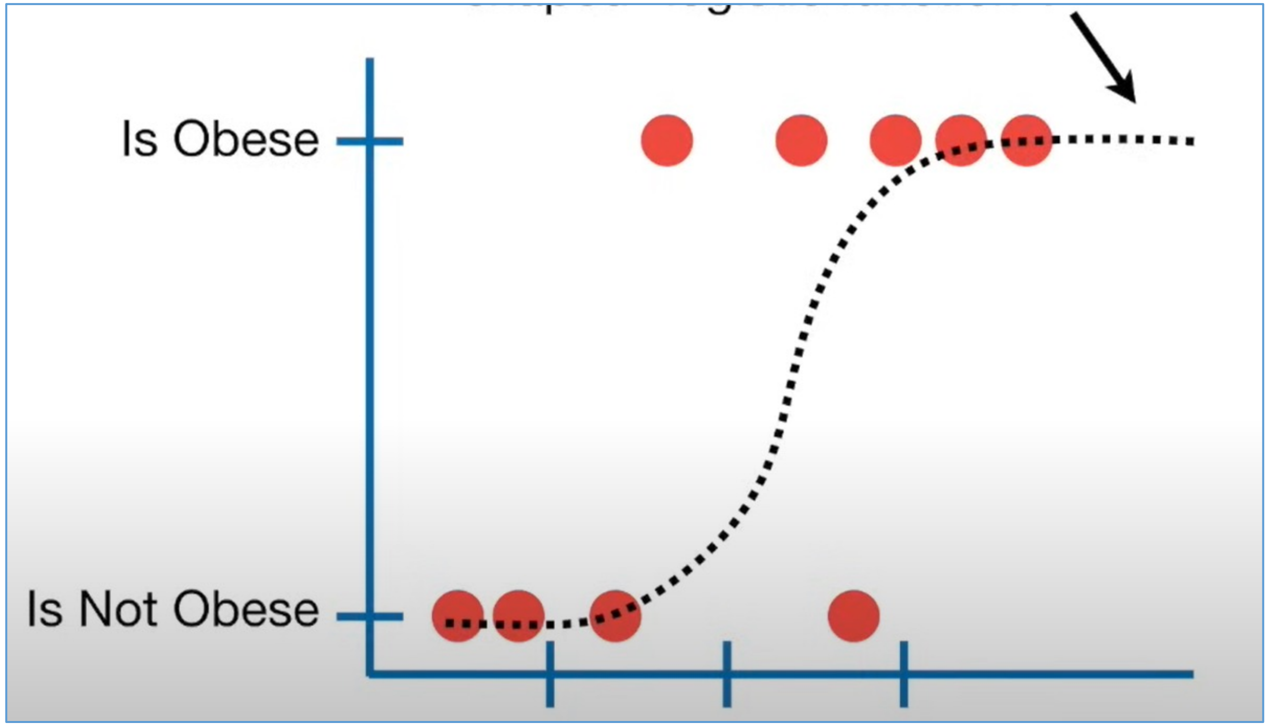
- Logistic regression is similar to linear regression except that Logistic regression **predicts whether something is true or False** instead predicting something continuous (e.g., predicting salary based on years of experience).

- In **logistic regression**, the goal is to **predict a probability** which is then converted into a binary outcome (e.g., spam or not spam).

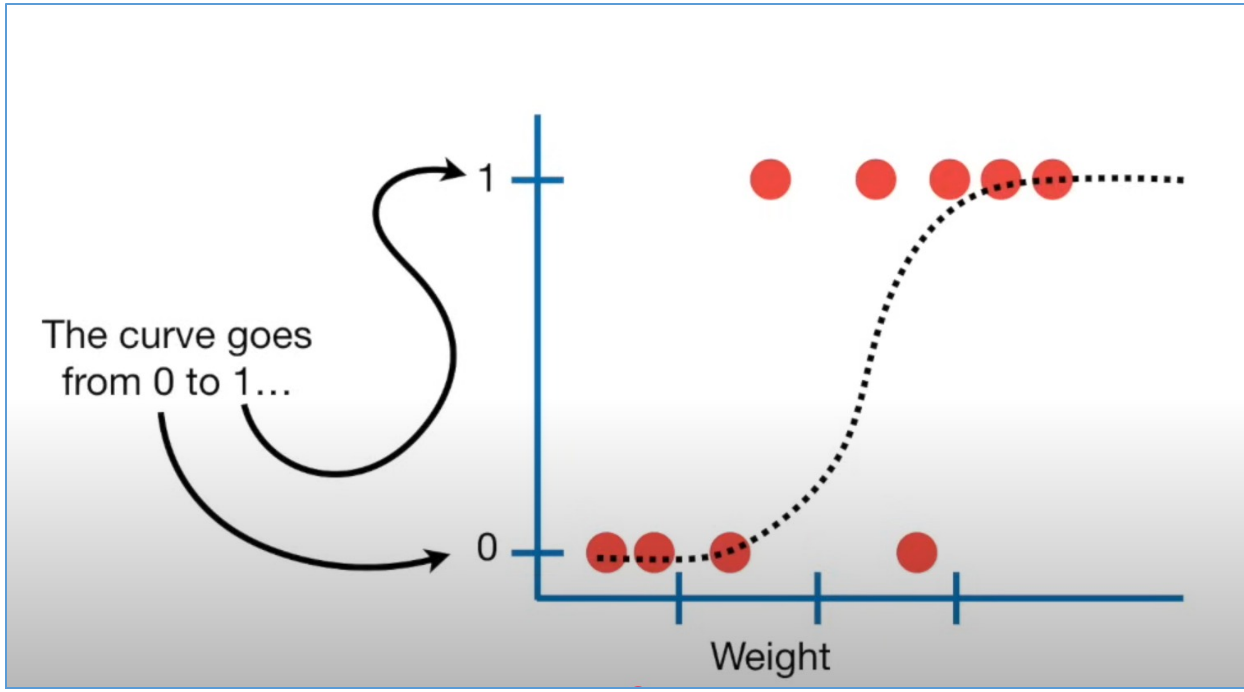
Logistic regression predicts whether something is true or False



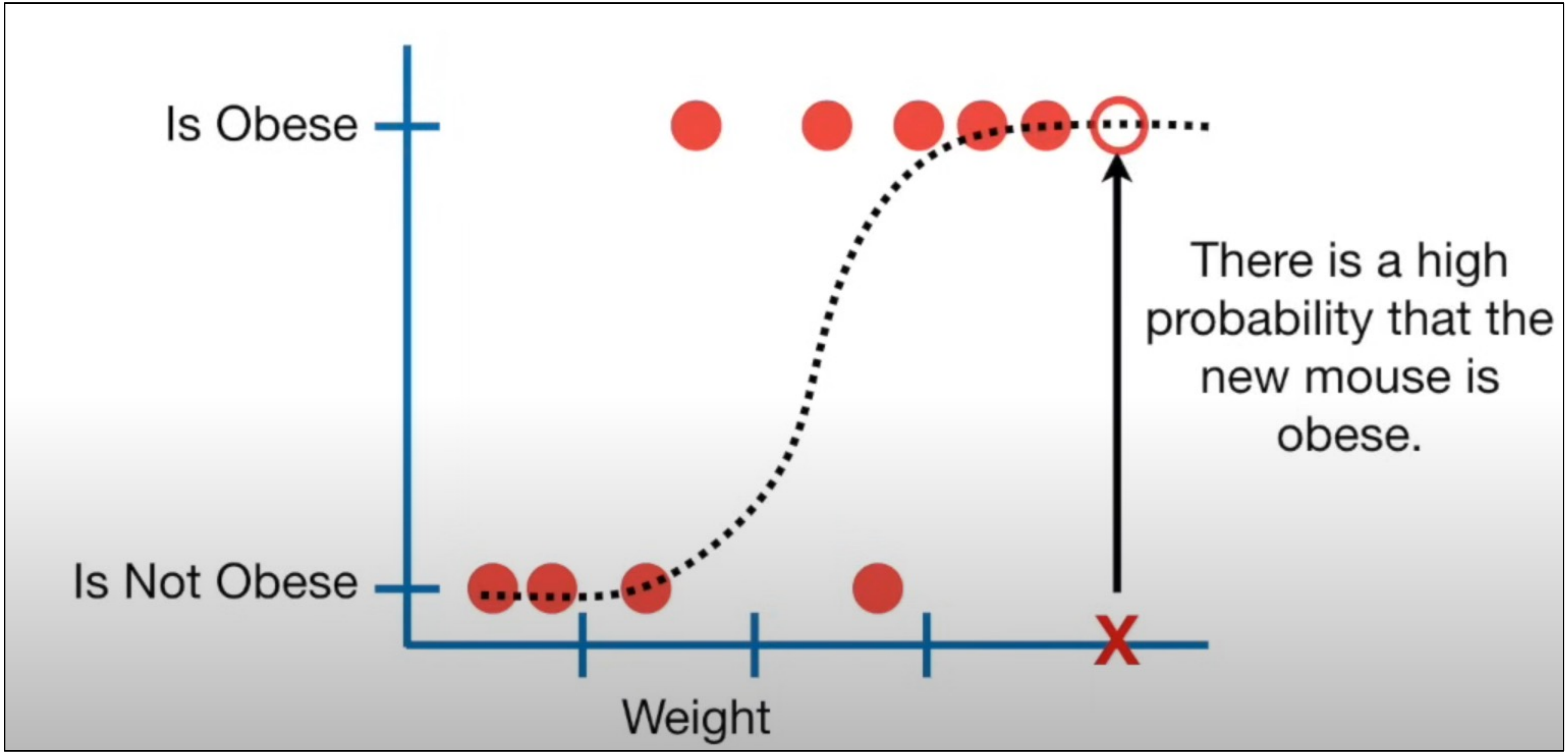
Instead of fitting a line to the data, logistic regression fits an "S" Shaped "Logistic Function"



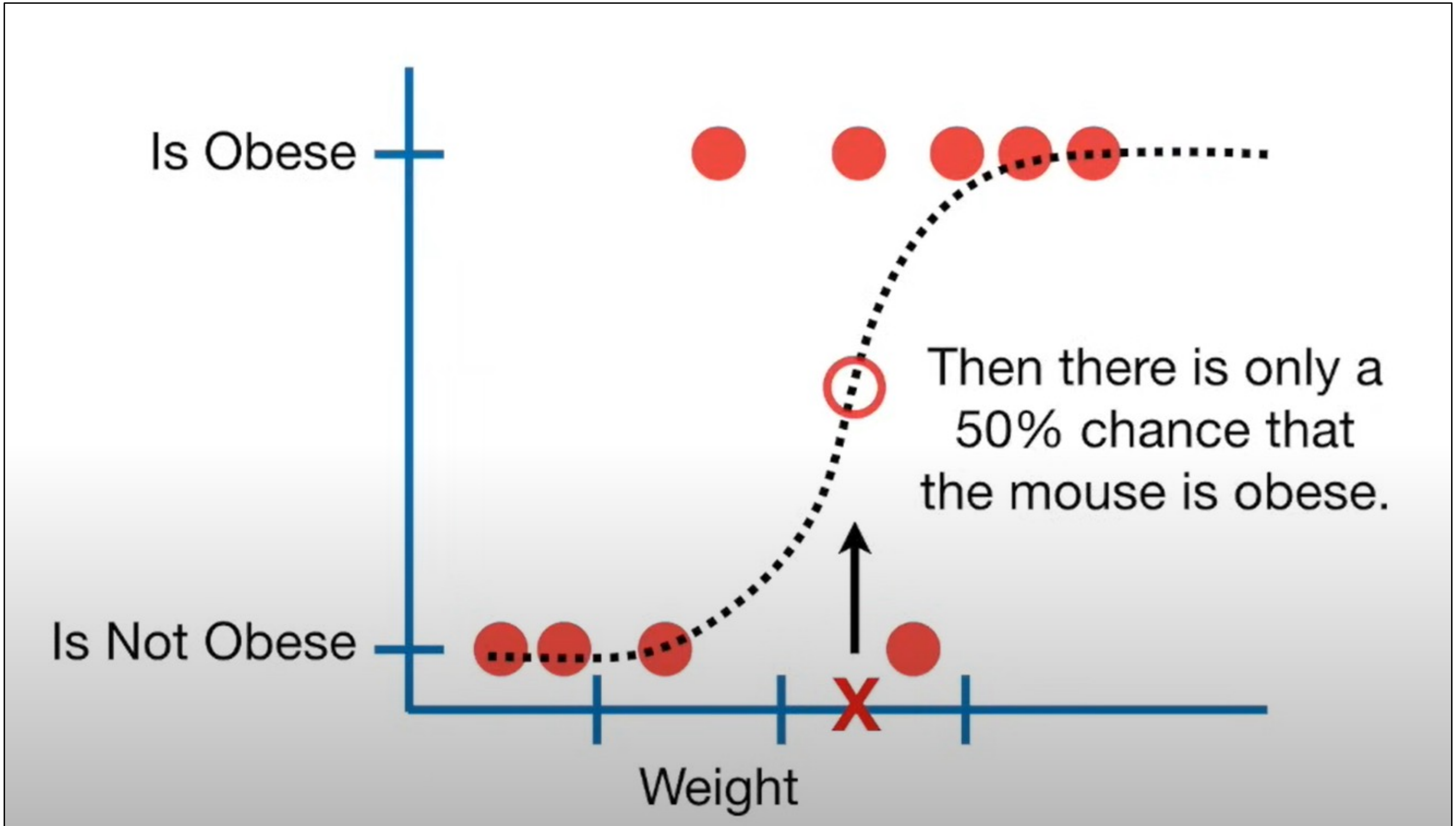
- The curve goes from o to 1
- The **curve tells you the probability** that mouse is Obese based on its weight

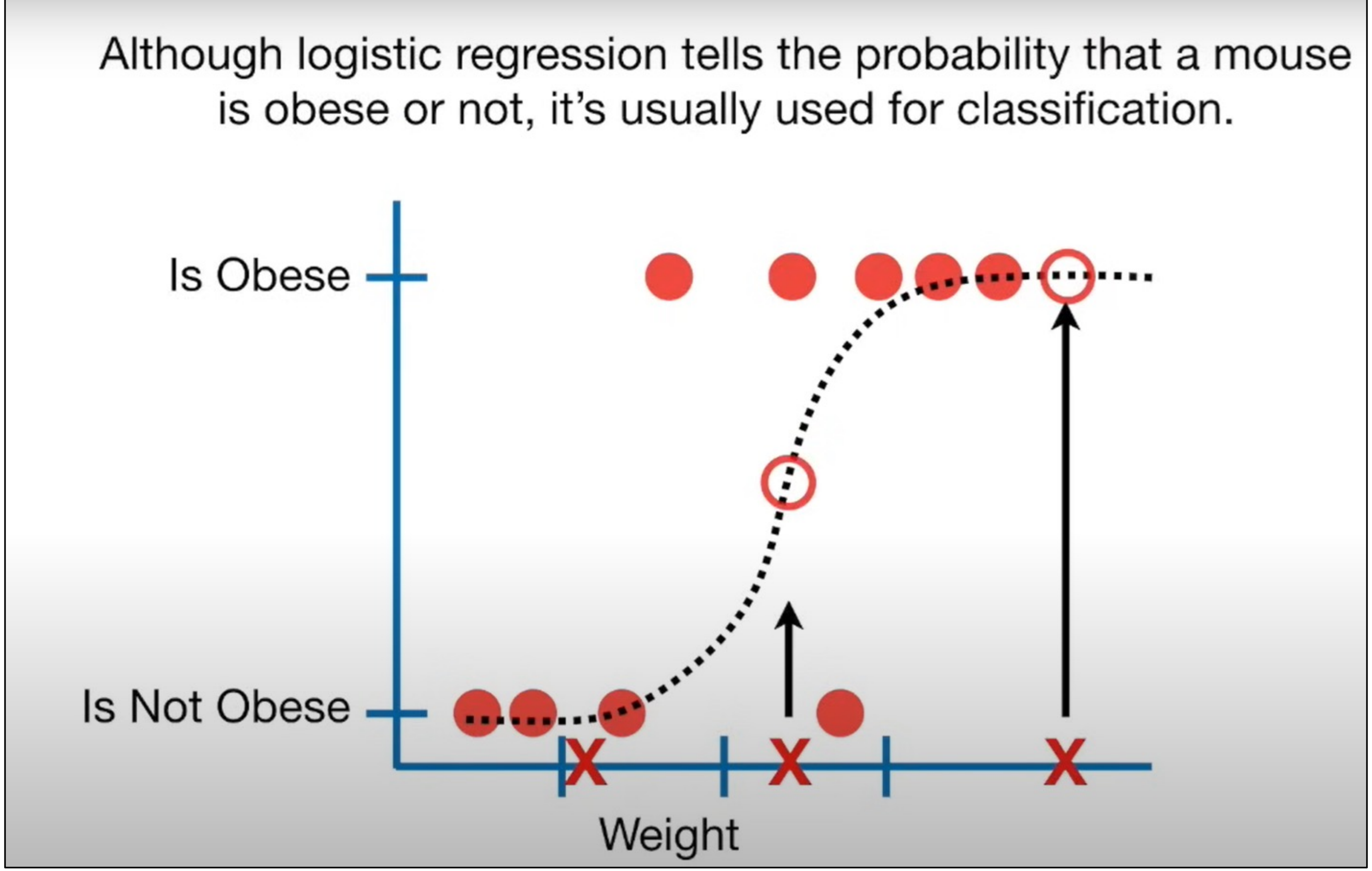


If it weighted a heavy mouse, then there is high probability that the new mouse is obese



If we weighted an intermediate mouse then there is only 50% chance that the mouse is obese





For Example, if the probability a mouse is obese is >50%, then we will classify it as "obese", Otherwise we will classify it as "not obese"

Logistic Regression

- Logistic regression models the **log-odds** (or **logit**) of the outcome as a **linear combination** of the independent variables:

$$\log \frac{P(Y=1)}{P(Y=0)} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$$

where:

- $P(Y=1)$ = Probability that the outcome is 1.
- $\beta_0, \beta_1, \ldots, \beta_n$ = Coefficients learned from the data.
- $X_1, X_2, \ldots, X_n$ = Independent variables.

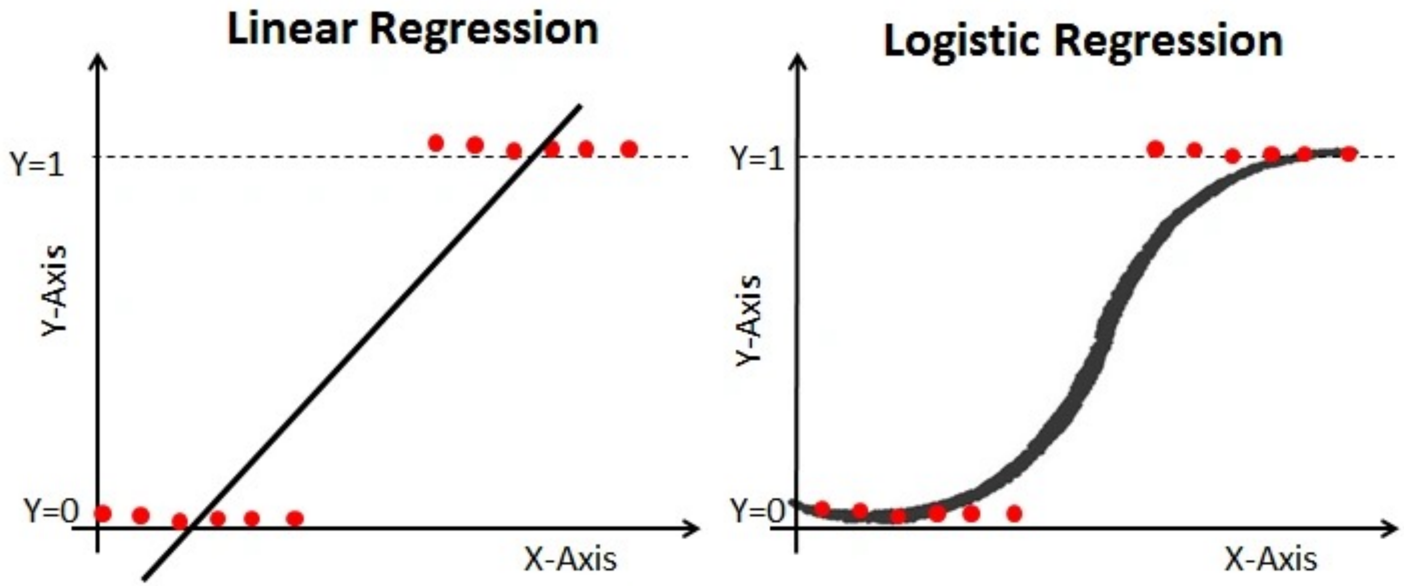
The predicted probability is obtained by applying the **sigmoid function**:

$$P(Y=1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n)}}$$

Formula for Logistic Function

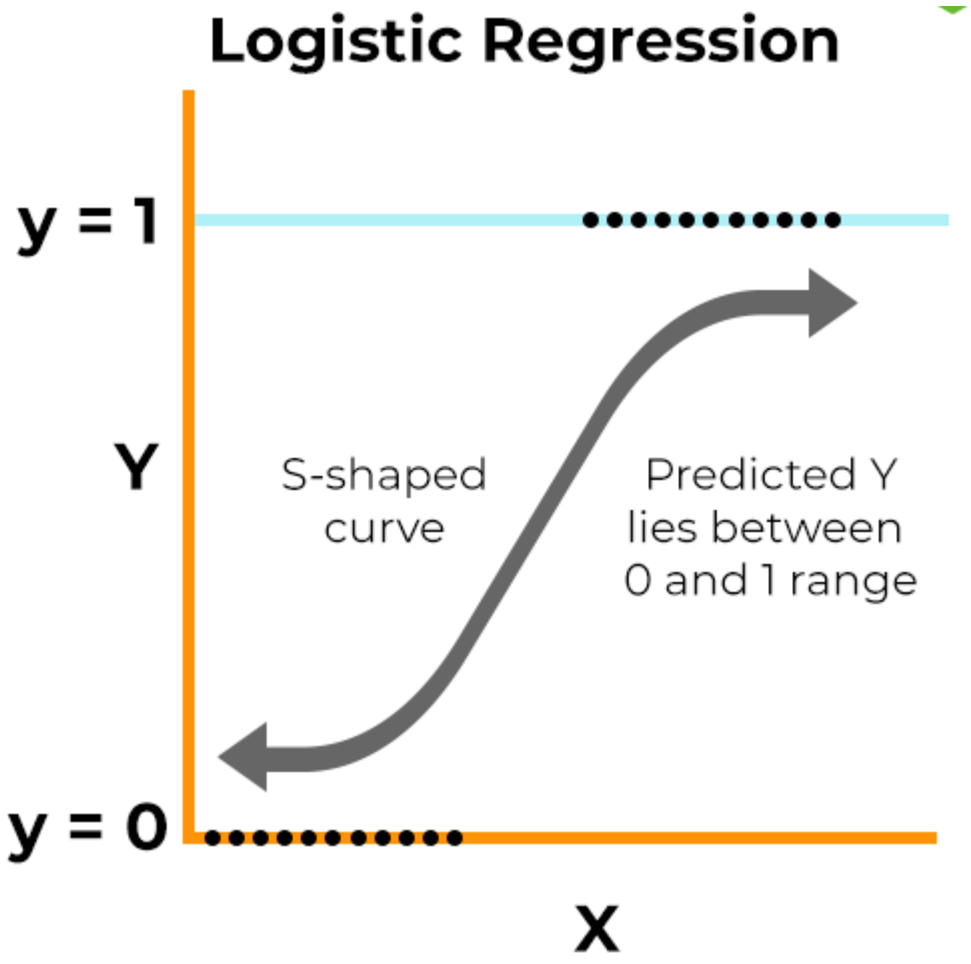
$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Logistic function, also called a sigmoid function.



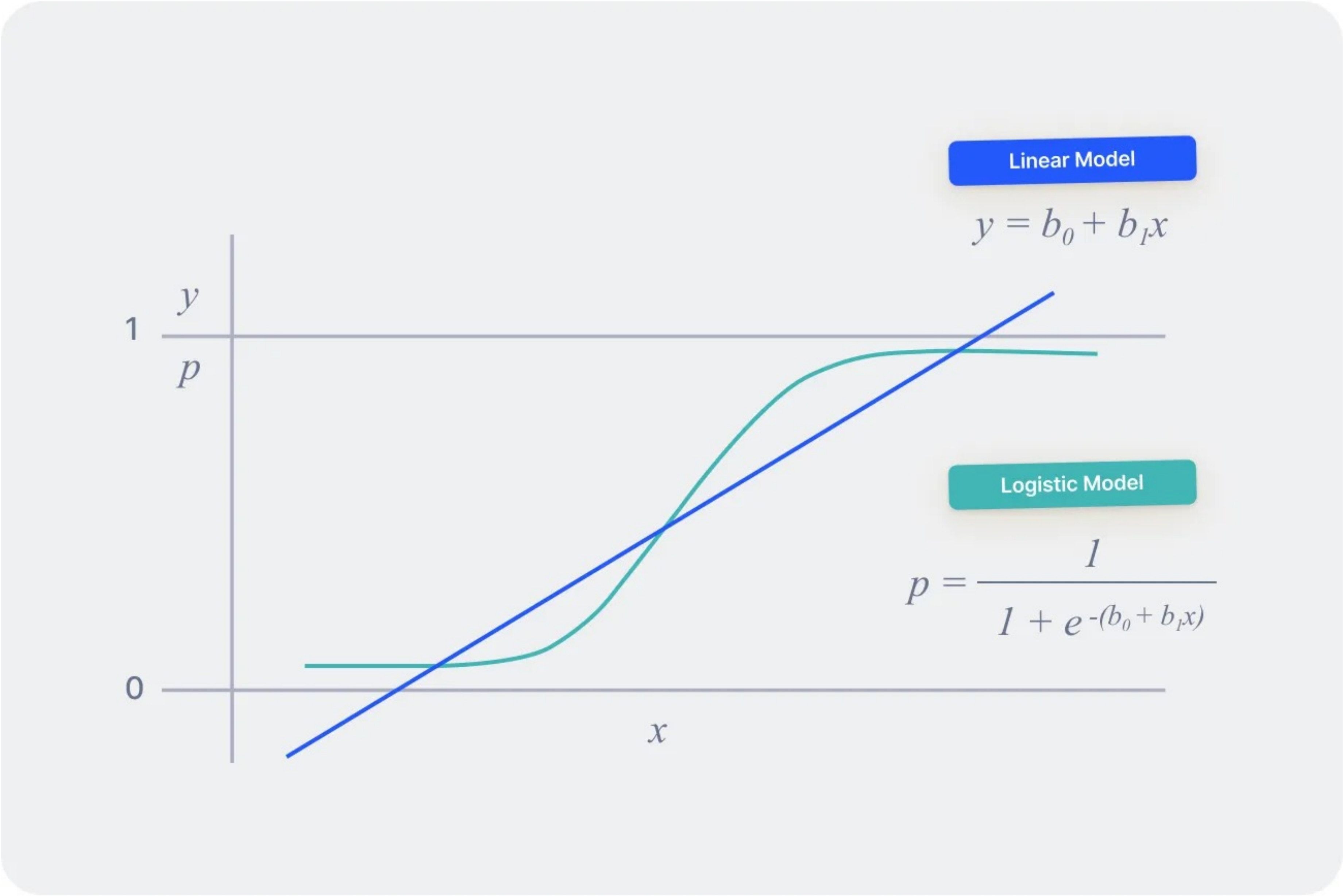
This function squeezes a straight line into an S-curve.

Sigmoid function



- Logistic regression uses a logistic function called a **sigmoid function to map the predicated linear combination of independent variables to probability between 0 and 1**

- The sigmoid function refers to an S-shaped curve that converts any real value to a range between 0 and 1.


Linear Model
y = b0 + b1x
Logistic Model
p = 1 / (1 + e^-(b0 + b1x))
y
1
p
0
x

Example of Logistic Regression Converting a Linear Combination to Probability

Let's say we want to predict whether a student will **pass** or **fail** an exam based on the following input features

- X1= Number of study hours
- X2 = Number of practice tests taken

Step 1: Define the Logistic Regression Model

$$Z=\beta_0+\beta_1 X1+\beta_2 X2$$

where:

- Z = Linear combination of inputs
- β_0 = Intercept (bias)
- β_1, β_2 = Coefficients for study hours and practice tests

Suppose the learned coefficients are:

- $\beta_0 = -2$
- $\beta_1 = 0.8$
- $\beta_2 = 0.5$

Step 2: Compute the Linear Combination

If a student studies for **5 hours** and takes **3 practice tests**, the linear combination becomes:

$$Z = -2 + (0.8 \times 5) + (0.5 \times 3)$$

$$Z = -2 + 4 + 1.5 = 3.5$$

Step 3: Convert to Probability using Sigmoid Function

The probability is computed using the **sigmoid function**:

$$P(Y=1) = \frac{1}{1+e^{-Z}}$$

Substituting $Z = 3.5$:

$$P(Y=1) = \frac{1}{1+e^{-3.5}}$$

$$P(Y=1) = \frac{1}{1+0.0302} = \frac{1}{1.0302} \approx 0.9707$$

The model predicts that the student has a 97.07% probability of passing the exam based on 5 study hours and 3 practice tests.

Reference

- https://www.youtube.com/watch?v=yIYKR4sgzI8&t=189s